

Change evidence report

CONSTRUCTION

Waiting for review

Construction appeared by **18 Dec 2024**, covering about 4.3 hectares.

AREA

Highland Sector

WHAT HAPPENED

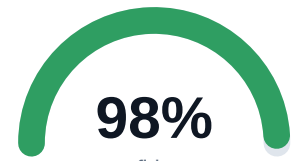
Appeared

LOCATION

34.0974°N, 75.0116°E

SEEN BETWEEN

18 Nov 2024 and 18 Dec 2024

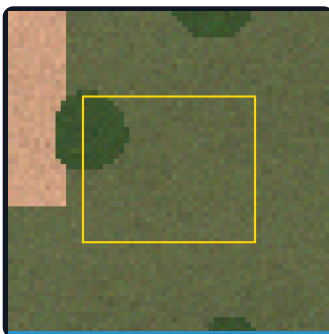


confidence

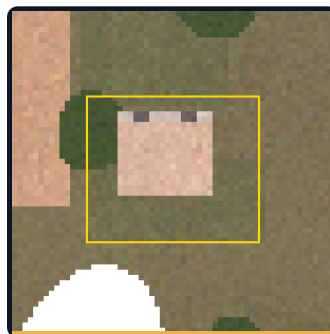
Very likely real

What the satellite saw

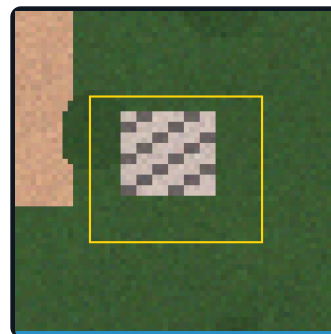
yellow box marks the changed area



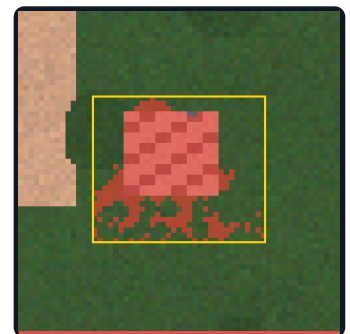
Before
18 Nov 2024



First sign
18 Dec 2024



Latest
6 Sep 2025



Changed area
highlighted in red

What the ground was and what it became

BEFORE | 18 NOV 2024

100%

WHEN DETECTED | 6 SEP 2025

32%

59%

Vegetation Built-up Roads Earthworks

Why SAMVID trusts this alert

0 = weak, 100 = strong

Strength of the change

68

far outside normal seasonal variation

Image clarity

100

no cloud or shadow over the area

Image alignment

92

off by 0.12 pixels at most

Shape consistency

86

changed pixels form one solid patch

Radar check

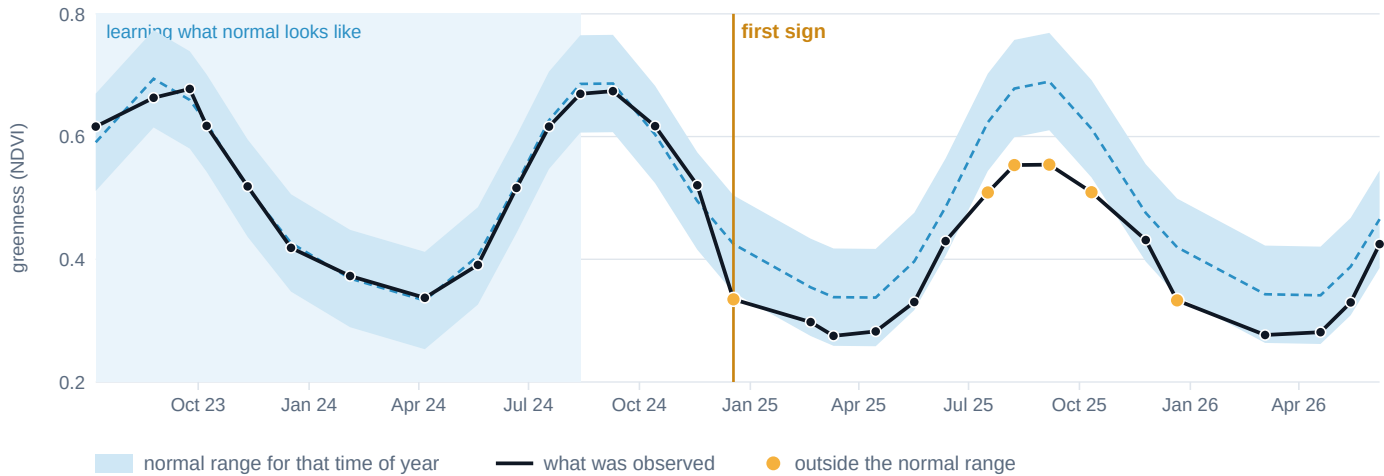
95

radar sees the same change

Seasonal check

is this just the time of year?

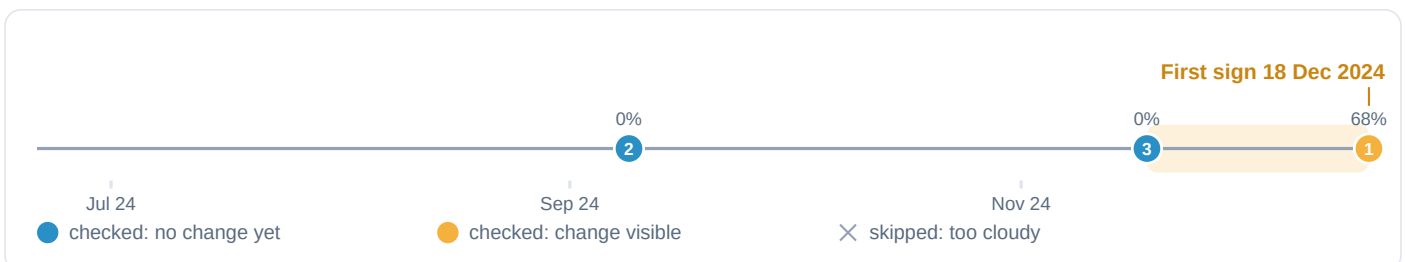
Fields green up and dry out every year, so SAMVID first learns the normal greenness of this exact spot for each month. The shaded band is that normal range. Points that fall outside it cannot be explained by the season. At detection, greenness was **0.201** where **0.5** was expected for that time of year.



Finding the first sign

3 images checked instead of 6

Instead of opening every image in order, SAMVID jumps to the middle of the time window, checks for the change, and halves the window each time. Cloudy images are skipped. Numbers show the order of the checks; percentages show how much of the changed area was visible.



Second opinion and decision

Radar check (Sentinel-1)

Agrees

Radar backscatter changed by **-3.46 dB** between 11 Sep 2024 and 6 Jan 2025. Radar sees through cloud, so it is an independent check.

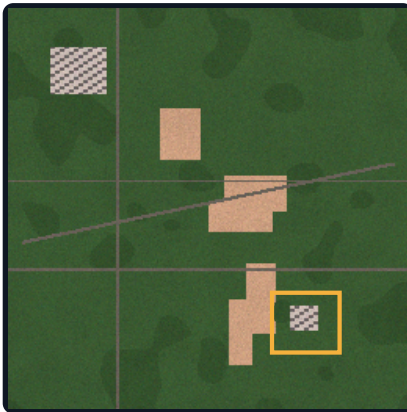
Analyst decision

Waiting for review

No one has reviewed this alert yet.

Where it is

full scene at detection



AREA

Highland Sector

CENTRE

34.09743 N, 75.01162 E

SIZE ON THE GROUND

4.3 hectares

IMAGE USED

S2_AOI-HIL_20250906

GRID TILES

5:5, 5:6, 6:5, 6:6

How this alert was produced

Step	What happened	Method / version
1. Ingest + QC	scenes 15..26 cloud/shadow/snow masked, co-registered, PIF-normalised	rule-masks v1.2 + phase-correlation coreg + PIF normalisation
2. Seasonal normalisation	anomalies vs per-pixel harmonic baseline	robust 2-harmonic baseline v1.0
3. Change detection	4 detection(s) over quarterly pairs, z>3.0	anomaly z-score + HDBSCAN(min_cluster_size=12) v1.1
4. Earliest observation	3 comparisons (linear scan: 6)	binary-search earliest-observation v1.0
5. SAR corroboration	-3.46 dB change, agrees	log-ratio VV

Chain of custody

tamper-evident audit ledger



Ledger intact

All 301 entries re-checked when this report was made.

head 246234283f9e0818cb6846573d8ff137d258312c2d873e837aac8cb90dcf1fee

#	Date	By	Action	Entry fingerprint (SHA-256)
#223	3 Oct 2026	system:monitor	Alert created	ab193557e98f9869caee15129235bcc7ef0d32c4ed6bff0a8323224691f5a7fc
#290	4 Oct 2026	analyst	Report exported	fbf88e75762665da1db9431dc1b80c8b9b4c9a5b7a5576a65af31f9afac3a440
#294	4 Oct 2026	analyst	Report exported	4bede2be5cbcec3947f41562c6348868a112cfea972f79df4194cc5e1d3b9c23
#297	4 Oct 2026	analyst	Report exported	c5ff1c8d719111fd3ce79e3a9d2a1fd9d5bdf6c956a3f21c6162281ff7f66222
#299	4 Oct 2026	analyst	Report exported	73420d3b42acd5a895fe1432bc2fef1cec2fc44bb78906d8d928a44dbd096bab
#302	4 Oct 2026	analyst	Report exported	246234283f9e0818cb6846573d8ff137d258312c2d873e837aac8cb90dcf1fee

Each entry's fingerprint is computed from its contents plus the previous entry's fingerprint, so changing any past entry breaks every one after it. Export the full ledger from the Audit trail page to check this independently.